

Paper Club · Sep 9

Archaeologist

11 + 3 minutes

Lineage · priors · subsequent work

arXiv 2512.06951 · 1st Place BEHAVIOR 2025

Seat 2

Lineage Map · Where this entry sits

Benchmarks / Tasks

BEHAVIOR-1K

OmniGibson / BDDL

VLA backbone

π_0 / $\pi_{0.5}$

SigLIP / PaliGemma

RT-1 / RT-2

GR00T (foil)

Action models

Flow Matching

Diffusion Policy

FAST tokenizer

Chunking / control

ACT chunking

RTC / real-time chunking

Hi Robot (foil)

This work (Dec 2025)

Correlated FM

System 2 vote

Soft inpaint

Gripper recovery

Subsequent: too new — author repo / blog / video only so far

π_0 / $\pi_{0.5}$ lineage

OpenPI flow-matching VLA → task-adapted contest entry

Motion / Control Spine · Not classical planning

Learned action-chunk FM + rolling receding-horizon + stage hierarchy + rule recovery



Closed-loop overrides (outer loop)

System 2 · Stage Voting

5-15 stages / task
vote window = 3 ($\geq 2/3$ forward)
rollback rules for aliasing

NON-MARKOVIAN context

Gripper Recovery Heuristic

force open if closed in stage
never closed in demos
paper: $\sim 2.2 \times q$ on 13×3 eps

CLOSED-LOOP OVERRIDE

Foils & contrasts

GR00T · Hi Robot · Diffusion Policy · RT-1/2

Subsequent work

Dec 2025 · repo / blog / video only so far

Takeaway

$\pi_{0.5}$ + correlated FM + RTC-style chunking + stage vote + heuristics